

Short Papers

Kinematic Design of a Six-DOF Parallel-Kinematics Machine With Decoupled-Motion Architecture

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Abstract—The design of a new six-degree-of-freedom (6-DOF) parallel-kinematics machine (PKM) has been proposed. Different from the conventional Stewart–Gough platform which has six extensible legs, the new PKM employs three identical RPRS legs to support the moving platform. Since all joint axes, excluding the three spherical joints at the leg ends, are parallel to each other and perpendicular to the base plane, this 6-DOF PKM presents a promising platform structure with decoupled-motion architecture (DMA) such that translation in a horizontal plane and rotation about a vertical axis are driven by the three active revolute joints, while translation in the vertical direction and rotation about horizontal axes are driven by the three active prismatic joints. As a result, this 6-DOF 3RPRS PKM with DMA has simple kinematics, large cylindrical reachable workspace, and high stiffness in the vertical direction. These features make it appropriate for light machining and heavy parts assembly tasks. Because of the DMA, a projection technique is employed for its kinematics analysis. By projecting the manipulator onto horizontal directions and vertical planes, the kinematics issues such as the displacement, singularity, and workspace analysis are significantly simplified.

Index Terms—Configuration design, decoupled-motion architecture (DMA), kinematics analysis, parallel-kinematics machine (PKM).

I. INTRODUCTION

A parallel-kinematics machine (PKM) is a closed-loop mechanism in which the moving platform is connected to the base by at least two serial kinematic chains (legs). The conventional Stewart–Gough platform has six extensible legs and hence leads to a rigid kinematic structure. However, such a six-legged fully parallel manipulator has coupled-motion characteristics, complicated kinematics, and limited workspace, which make its design and application development difficult. The PKMs with a fewer number of legs, decoupled motions, simple kinematics, and large workspace have been received a great attention in recent robotics literature, e.g., the three-legged 3-degrees-of-freedom (DOF) Universal Cartesian Robot (UCR) [1], [2], the three-legged 6-DOF Tri-Scott manipulator [4], and the three-legged 6-DOF Eclipse manipulator [5]. It is noted that a PKM is considered having decoupled-motion architecture (DMA) if some of its primary motions (e.g., translation along with or rotation about an axis) are always driven by certain (not all) active joints. A PKM with DMA also implies that the PKM should have simple kinematics so as to facilitate its analysis, design, and motion control.

In this paper, a 6-DOF PKM with DMA is proposed for heavy parts assembly and light machining (e.g., surface finishing) tasks. As

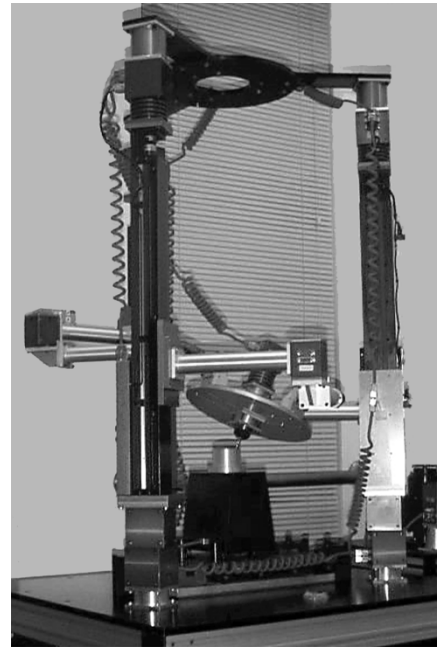


Fig. 1. Prototype of the 6-DOF 3RPRS modular PKM with DMA.

shown in Fig. 1, this PKM consists of three identical SCARA-like RPRS (R-Revolute joint, P-Prismatic joint, and S-Spherical joint) legs supporting the moving platform. Excluding the spherical joint at each of the leg ends, each leg forms a RPR serial kinematic chain with all joint axes parallel to each other and perpendicular to the base plane. Specifically, the first and second joints in each leg are active joints and their motion axes are collinear. As a result, such a PKM inherits some basic characteristics of a SCARA-type serial robot such as decoupled motions, simple kinematics, high dexterity in the horizontal plane, and high stiffness in the vertical direction. Nevertheless, its overall structural rigidity is higher than that of its serial counterpart because of its parallel structure.

In order to shorten the development cycle, the modular design concept is employed in the prototype development. The modularly designed PKM system consists of a collection of standardized active and passive joint modules and customized links, connectors, and moving platforms. A modular PKM configuration thus can be rapidly constructed and its performance can be readily adjusted by changing the leg mounting positions, link lengths, joint types, and actuation schemes for a diversity of task requirements [6]. This design methodology is able to reduce the complexity of the overall design problem to a manageable level.

Another major concern of this paper is to investigate and formulate effective kinematics algorithms for the specific 6-DOF 3RPRS PKM with DMA, such as displacement, singularity, and workspace analysis. These algorithms are essential for the performance analysis and design optimization of the PKM.

A number of displacement analysis approaches have been proposed for the three-legged nonredundant PKMs. Innocenti and Castelli [7] proposed an analytical method for the forward displacement analysis of such PKMs, which led to solving a 16th-order polynomial equation. Consequently, a total of 16 sets of forwarded displacement solutions

Manuscript received February 24, 2004. This paper was recommended by Associate Editor J.-P. Merlet and Editor I. Walker upon evaluation of the reviewers' comments. This work was supported in part by the Singapore Institute of Manufacturing Technology (SIMTech), Singapore, under the Applied Research Project Grant C01-137-AR and in part by the University Collaborative Project under Grant U97-A-006. This paper was presented in part at the IEEE/ASME International Conference on Advanced Intelligent Mechatronics, Kobe, Japan, July 20–24, 2003.

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Digital Object Identifier 10.1109/TRO.2004.829485

were found out. Utilizing the sensed displacements of some passive joints, Notash and Podhorodeski [8] introduced a simplified analytical method for on-line computation of the moving platform poses. To avoid the complexity in solving high-order polynomial equations, Cleary and Brook [9] proposed a numerical method for the forward displacement analysis in which only one solution could be determined each time.

One of the early works to address configuration singularities of general closed-loop mechanisms is that of Gosselin and Angeles [10], in which three types of configuration singularities were classified. Hunt [11] laid down a general framework for applying screw theory to singularity analysis and introduced the notion of stationary and uncertainty configurations. Kumar [12] provided a more detailed analysis of closed-chain singularities by using the concept of screw reciprocity. Merlet [13] showed that by applying line geometry it is possible to enumerate all the singularities of the Stewart–Gough platform. Angeles [14], [15] introduced the concept of screw system annihilators to reduce the number of unknowns in the derivation of instantaneous kinematics model. Park and Kim [16] proposed a differential geometric analysis of singularities for closed-loop chains. A passive-joint velocity approach was proposed by Yang *et al.* [17] for a class of three-legged PKMs. Although most of these approaches can be employed here for singularity analysis, it is realized that they are not so efficient for this particular PKM configuration.

A 6-DOF PKM, in general, has a six-dimensional (6-D) (position and orientation) workspace. As it is very difficult to present the complete 6-D workspace in a human readable way, different subsets of the complete workspace are usually determined such as the constant-orientation workspace [18], the projected orientation workspace, and the orientation workspace [19]. A simple and generic approach to determine these workspace subsets is to use a discretization method [20]–[23]. However, such a numerical analysis method is computationally intensive and gives little information about the exact boundary of the workspace. An advanced geometric approach was first introduced by Gosselin [18], where boundaries of the horizontal cross sections of the workspace were determined. Merlet [24] extended this geometric approach by considering the joint limitations and leg interference. Different from the discretization methods, the geometrical approach is accurate and computationally efficient. Furthermore, it can give a parametric relationship between the design parameters and the workspace, which is of great importance during the design stage.

In this paper, based on the DMA of the 3RPRS PKM, a projection method is employed in order to simplify the kinematic analysis. By projecting the 3RPRS PKM onto the base (horizontal) plane, an imaginary 3RRR planar PKM can be obtained, which is able to reflect the kinematics of the 3RPRS manipulator in the horizontal plane. On the other hand, the linear motions of the three prismatic joints can be directly determined when projecting the 3RPRS PKM onto a vertical direction. Using such a projection technique, the displacement, singularity and workspace analysis of the 3RPRS PKM with DMA can be significantly simplified. As a result, the proposed kinematic analysis algorithms are computationally efficient.

The remaining sections of this paper are organized as follows. In Section II, the design of the new 6-DOF 3RPRS PKM with DMA is discussed. The displacement, singularity, and workspace analysis issues are then addressed in Sections III–V, respectively. The paper is summarized in Section VI.

II. DESIGN CONSIDERATIONS

A. Type Synthesis

In this section, we focus on the conceptual design of a 6-DOF nonredundant PKM for heavy parts assembly and light machining tasks that

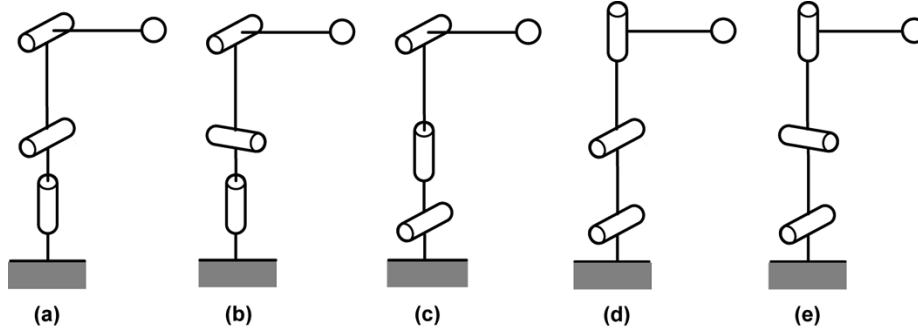
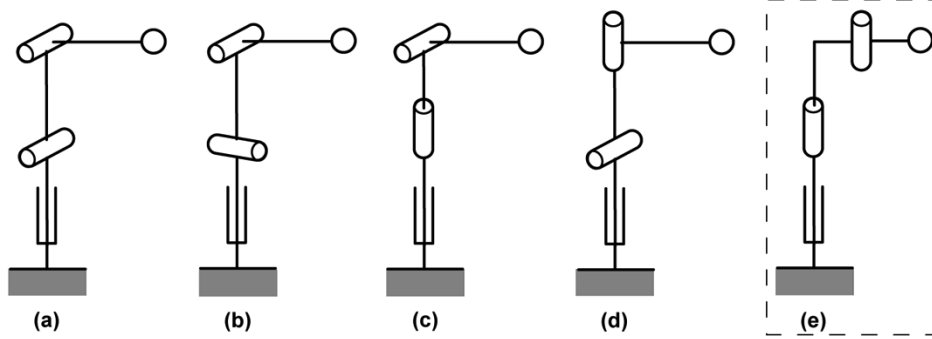
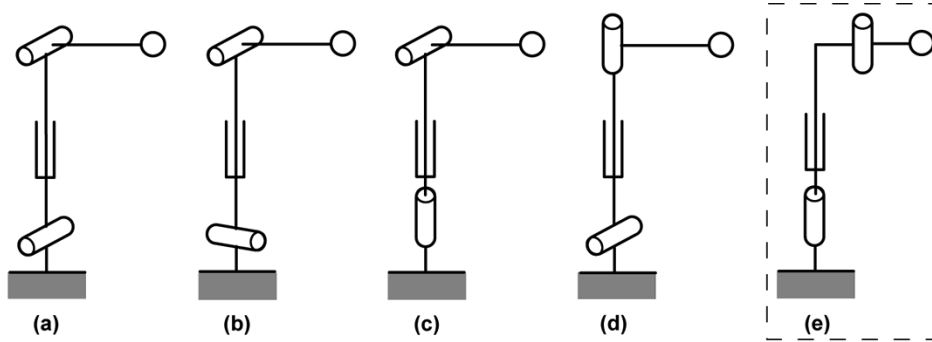
require large workspace, high dexterity, high loading capacity, and considerable stiffness, especially in the vertical direction. Kinematically, a 6-DOF nonredundant PKM also implies that each leg should also be a 6-DOF serial kinematic chain, regardless of the number of legs. With such a necessary condition, a topological structure enumeration scheme was proposed in [25] for 6-DOF PKMs, in which all possible PKMs with any number of legs between 2–6 were enumerated. Here, to simplify the design and development efforts, we have the following additional considerations.

- Symmetric design—each leg is identical to the others. Hence, each leg should have the same number of active joints. As the total number of (1-DOF) active joints in a 6-DOF nonredundant PKM is six, the number of legs for a symmetric design can be six, three, or two.
- Types of joints—four types of commonly used joints are being considered, i.e., 1-DOF revolute (R), 1-DOF prismatic (P), 2-DOF universal (U), and 3-DOF spherical (S) joints. Among them, the spherical and universal joints are only meant for passive joints, the prismatic joints are only meant for active joints (ineffective as passive joints), and revolute joints can be used as either passive or active joints.
- Active joints are placed close to the base so as to reduce the moment of inertia and increase the loading capacity and motion acceleration.
- A passive 3-DOF spherical joint is placed at each of the leg ends so as to reduce the number of passive joints and make the design compact.
- At most one (active) prismatic joint can be employed in each of the legs due to its heavy and bulky mechanical structure.

With these design considerations, we can enumerate all of the distinct leg structures for the 6-DOF symmetric PKMs with six, three, and two legs. In this design, however, we overlook the six- and two-legged PKMs because they are not as appropriate for the given tasks. The six-legged PKM has limited workspace and complex singularities, while the two-legged PKM has low stiffness and low loading capacity. Consequently, only the three-legged PKMs are considered for the new 6-DOF PKM design. As discussed in [6], [25], such three-legged 6-DOF PKMs have simple kinematics, larger workspace, and considerable loading capacity and stiffness, which make them very promising candidates for the given tasks.

Three possible leg topologies have been considered, i.e., RRRS, PRRS, and RPRS types, in which the underlined symbols represent active joints. For each of the leg topologies, five feasible leg structures are enumerated based on the joint orientations, as illustrated in Figs. 2–4. Among these fifteen feasible leg structures, there are two interesting SCARA-like leg structures, i.e., the PRRS leg shown in Fig. 3(e) and the RPRS leg shown in Fig. 4(e). Kinematically, these two legs are equivalent regardless of their different assembly sequences of the active joints. The unique feature of these two legs is that all joint axes in a leg (excluding the spherical joint at the leg-end) are parallel to each other and perpendicular to the base plane. As a result, the motions of the leg-end point in the horizontal plane and vertical direction are decoupled from each other such that the motion in a horizontal plane is only related to the two revolute joints and the motion in the vertical direction is solely driven by the prismatic joint. It follows that, with such a leg structure, it is possible to design a three-legged 6-DOF PKM with DMA for the given tasks.

Obviously, the two leg structures [Figs. 3(e) and 4(e)] are kinematically equivalent. In the prototype development, we selected the leg shown in Fig. 4(e), i.e., a RPRS leg, for ease of assembly and alignment. Through preliminary analysis, a 6-DOF PKM with such three RPRS legs (Fig. 7) possesses the following advantages:

Fig. 2. Feasible leg structures for RRRS leg.Fig. 3. Feasible leg structures for PRRS leg.Fig. 4. Feasible leg structures for RPRS leg.

- DMA—translation in a horizontal plane and rotation about a vertical axis ($x-y-\theta_z$) are driven by the three active revolute joints, while translation in the vertical direction and rotation about a horizontal axis ($z-\theta_x-\theta_y$) are driven by the three prismatic joints.
- Simple kinematics and easy for analysis, design, trajectory planning, and motion control.
- Simplex singularity configurations.
- Large and well-shaped cylindrical reachable workspace.
- High stiffness in the vertical direction.
- High loading capacity.

B. Robot Modules

For rapid deployment, the modular design concept is employed. A set of standardized active and passive joint modules and customized links, connectors, and moving plates are used as the basic modules for constructing the modular PKM. Off-the-shelf actuators (PowerCube from Amech GmbH, Germany) are selected for the active joint modules, i.e., the 1-DOF prismatic joint module PSM90 [Fig. 5(a)] and the 1-DOF revolute joint module PR90 [Fig. 5(b)]. These active joint

modules are self-contained and intelligent driving units with built-in motors, controllers, amplifiers, and communication interfaces. Each of them has a cube-like design with multiple connecting sockets so that any two joint modules can be connected through links and connectors in different orientations. The passive joint modules, e.g., the 1-DOF revolute joint module [Fig. 5(c)] and the 3-DOF spherical joint module [Fig. 5(d)], are of standard design, while the links, connectors, and moving plates are custom-designed. Note that the spherical joint module is modified from a universal joint by adding a rotary bearing in each of its two connecting ends. In total, there are four revolute joints in a modified universal joint module. Since these four joint axes intersect at the same point, the modified universal joint is functionally the same as a spherical joint, but having large rotation angles due to joint redundancy. Besides, in order to improve the structural rigidity of joints, a heavy-duty liner guide with double slide blocks is employed for a prismatic joint to sustain large bending torque, while two sets of cross-roller bearings are employed for a passive revolute joint to sustain large axial/radial force as well as a large bending torque.

Using the standardized joint modules and customized links and moving plates, a 6-DOF 3RPRS modular PKM with DMA can be

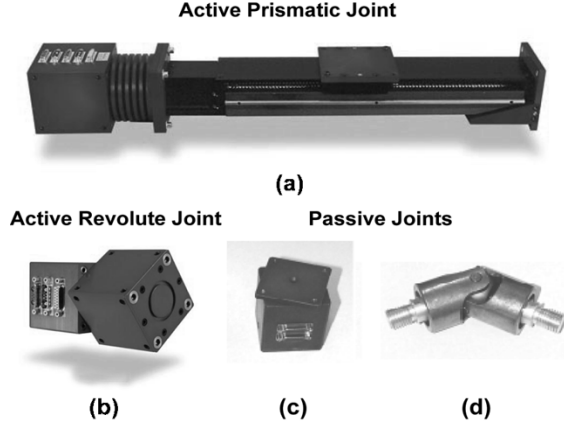


Fig. 5. Standardized joint modules. (a) 1-DOF prismatic joint module PSM90. (b) 1-DOF revolute joint module PR90. (c) 1-DOF revolute joint module. (d) 3-DOF spherical joint module.

rapidly deployed as shown in Fig. 1. Due to the strengthened structures of joints and links, the PKM has considerable stiffness, especially in the vertical direction. To improve the structural rigidity, three additional passive-revolute-joint modules are employed to connect the distal ends of the three prismatic joint modules to a triangular plate. It is apparent that this triangular plate can also be rigidly connected to the base to further enhance the rigidity of the system. Furthermore, the locations of the three additional passive-revolute-joint modules and those of the three active-revolute-joint modules can be exchanged without altering the kinematics of the PKM. When it happens, the reaction force of the three active-revolute-joint modules will be decreased as the weight of the PKM will be carried by the three additional passive-revolute-joint modules which are placed at the bottom. Therefore, in the newly modified design, the triangular plate is rigidly connected to the base through three supporting columns, while the three active-revolute-joint modules are placed on the top of the prismatic joint modules, as shown in Fig. 6.

III. DISPLACEMENT ANALYSIS

The purpose of displacement analysis is to derive the kinematic relationship between the active-joint displacements and the end-effector poses. The schematic diagram of the 3RPRS PKM with DMA is shown in Fig. 7. Each leg has an active-revolute joint $\hat{s}_{bi}$, an active-prismatic joint $\hat{s}_{ci}$, a passive-revolute joint $\hat{s}_{di}$, and a passive-spherical joint at leg-end point P_i ($i = 1, 2, 3$). A local coordinate frame is attached to each of the revolute and prismatic joints with the z axis coincident with its respective joint axis. Let frame B be the base frame and frame E be the end-effector frame which is attached to the moving platform. The coordinates of point P_i ($i = 1, 2, 3$) (centers of the spherical joints) with respect to frames B and E are represented by $p_i^B = (x_{pi}^B, y_{pi}^B, z_{pi}^B)$ and $p_i^E = (x_{pi}^E, y_{pi}^E, z_{pi}^E)$, respectively. When point P_i is expressed in local frame D_i on the third joint of leg i , its position vector is denoted as $p_i^{D_i} = (x_{pi}^{D_i}, y_{pi}^{D_i}, z_{pi}^{D_i})$.

A. Forward Displacement Analysis

The forward displacement analysis is to determine the pose of the end-effector frame E when the six active-joint displacements ($q_{b1}, q_{c1}, q_{b2}, q_{c2}, q_{b3}$, and q_{c3}) are known. For an open-chain mechanism, this is simple and straightforward because all joints are active and their displacements are known. For a parallel manipulator, however, it is usually difficult because of the existence of passive joints

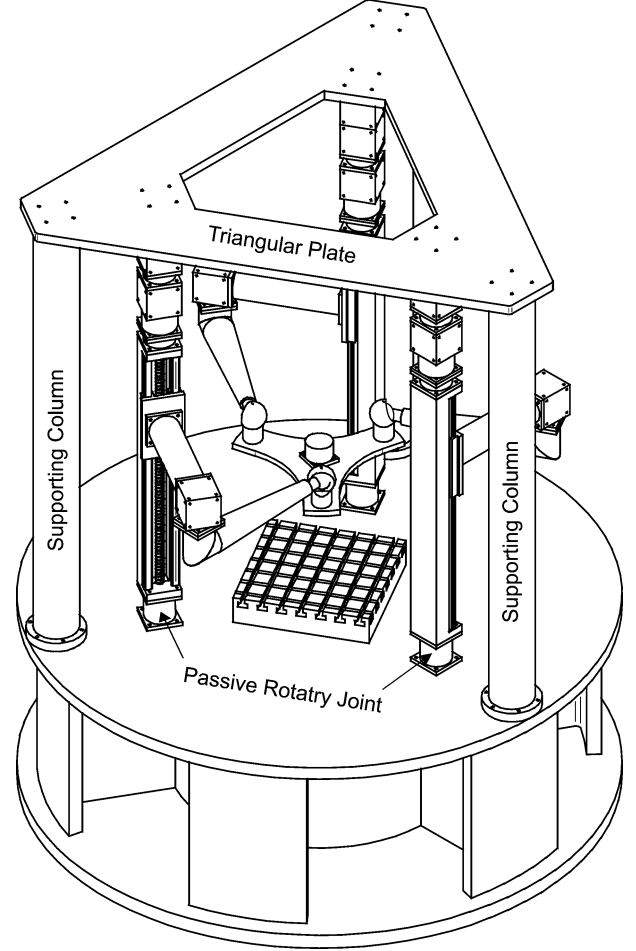


Fig. 6. Modified design of the 3RPRS PKM with DMA.

whose displacements are unknown. To perform the forward displacement analysis, in general, the passive-joint displacements have to be determined first with the given active-joint displacements.

Let the forward kinematic transformation from the base frame B to the end-effector frame E be $T_{B,E} \in SE(3)$, then

$$T_{B,E} \begin{bmatrix} p_i^E \\ 1 \end{bmatrix} = \begin{bmatrix} p_i^B \\ 1 \end{bmatrix}, \quad i = 1, 2, 3. \quad (1)$$

Since both frames B and E are right-hand Cartesian coordinate frames, we also have

$$T_{B,E} \begin{bmatrix} p_{12}^E \times p_{23}^E \\ 0 \end{bmatrix} = \begin{bmatrix} p_{12}^B \times p_{23}^B \\ 0 \end{bmatrix} \quad (2)$$

where $p_{12}^B = p_2^B - p_1^B$, $p_{23}^B = p_3^B - p_2^B$, $p_{12}^E = p_2^E - p_1^E$, and $p_{23}^E = p_3^E - p_2^E$. Combining (1) and (2), the pose of the end-effector frame, $T_{B,E}$, can be found by

$$T_{B,E} = \begin{bmatrix} p_1^B & p_2^B & p_3^B & p_{12}^B \times p_{23}^B \\ 1 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} p_1^E & p_2^E & p_3^E & p_{12}^E \times p_{23}^E \\ 1 & 1 & 1 & 0 \end{bmatrix}^{-1}. \quad (3)$$

Note that the matrix whose inverse appears in (3) is a constant matrix for a specific robot design and is always invertible if points P_1 , P_2 , and P_3 neither coincide nor are collinear so that $p_{12}^E \times p_{23}^E \neq 0$.

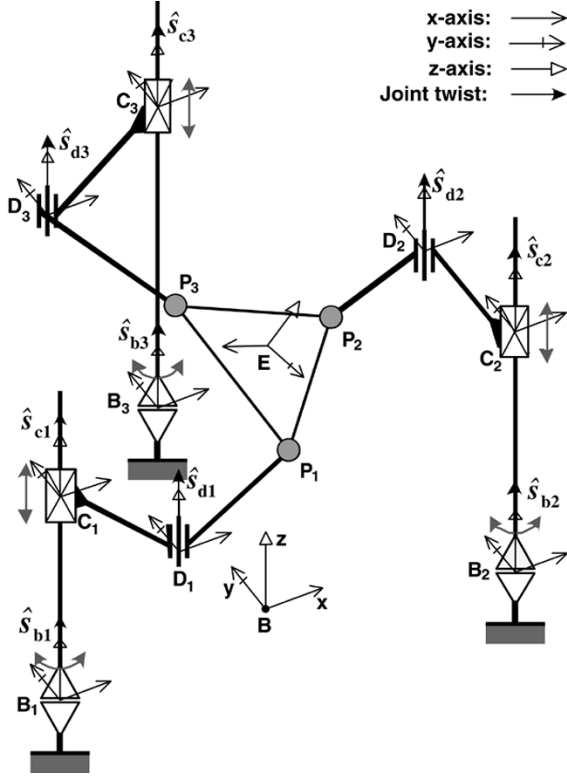


Fig. 7. Schematic drawing of the 3RPRS PKM with DMA.

Equation (3) indicates that $T_{B,E}$ can be readily determined if the position vector of point P_i ($i = 1, 2, 3$) with respect to the base frame, i.e., p_i^B , is known. Considering each of the individual legs in the 3RPRS PKM with DMA as shown in Fig. 7, p_i^B can be written as

$$\begin{bmatrix} p_i^B \\ 1 \end{bmatrix} = T_{B,B_i}(0) e^{\hat{s}_{bi} q_{bi}} T_{B_i,C_i}(0) e^{\hat{s}_{ci} q_{ci}} \cdot T_{C_i,D_i}(0) e^{\hat{s}_{di} q_{di}} \begin{bmatrix} p_i^D \\ 1 \end{bmatrix}. \quad (4)$$

Equation (4) is formulated based on the local frame representation of the *products-of-exponentials* (POE) formula [17], [26], [28], [27].

In (4), the only unknown parameter is the passive-joint displacement q_{di} ($i = 1, 2, 3$). Therefore, the key issue for the forward displacement analysis is to determine the passive-joint displacement q_{di} with a given set of active-joint displacements.

Based on the kinematic features of the 3RPRS PKM with DMA such that all joint axes, except for the three spherical joints at the leg ends, are parallel to each other and perpendicular to the base plane, a projection method can be used in order to simplify the analysis [3], [4]. By projecting this 3RPRS PKM with DMA onto the base plane, i.e., the x_B - y_B plane, a planar 3RRR PKM can be obtained as shown in Fig. 8, where the prismatic joints are temporarily ignored and the spherical joints are treated as revolute joints. Since links $C_i D_i$ and $D_i P_i$ ($i = 1, 2, 3$) in the original 3RPRS PKM (Fig. 7) are parallel to the base plane, their link lengths as well as their related revolute-joint displacements (q_{bi} and q_{ci}) will remain unchanged after projection. However, the three leg-end distances $P_1 P_2$, $P_2 P_3$, and $P_3 P_1$ will be changed into $P'_1 P'_2$, $P'_2 P'_3$, and $P'_3 P'_1$, respectively, which can be computed as

$$\|P'_1 P'_2\|^2 = \|P_1 P_2\|^2 - (z_{p2}^B - z_{p1}^B)^2 \quad (5)$$

$$\|P'_2 P'_3\|^2 = \|P_2 P_3\|^2 - (z_{p3}^B - z_{p2}^B)^2 \quad (6)$$

$$\|P'_3 P'_1\|^2 = \|P_3 P_1\|^2 - (z_{p1}^B - z_{p3}^B)^2 \quad (7)$$

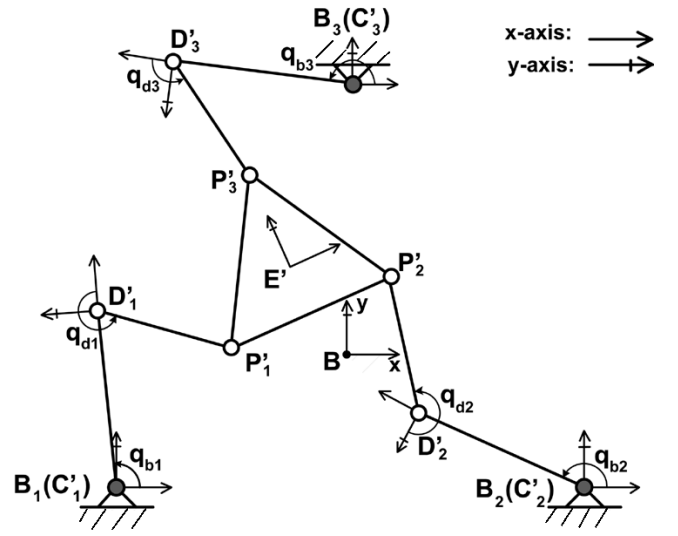


Fig. 8. Projected planar 3RRR PKM.

where z_{pi}^B ($i = 1, 2, 3$) represents the z coordinate of point P_i with respect to the base frame B , which can be uniquely determined by the prismatic joint displacement of leg i , i.e., q_{ci} . In other words, all geometrical dimensions of the projected planar 3RRR PKM can be uniquely determined.

The forward displacement analysis of such a general 3RRR planar PKM had been well studied by Gosselin and Sefrioui [29] in which a closed-form polynomial solution was derived. Since the closed-form solution leads to a polynomial of degree six, at most six real solutions exist for the projected 3RRR PKM. However, since the 3RRR PKM is a spatial mechanism, there are two possible triangular moving plates: the original plate and its mirror, satisfying the same three edge-length constraints [3]. Therefore, we can conclude that at most 12 real solutions exist for the forward displacement analysis of the 3RPRS PKM with DMA. For more details about this method, please refer to [3].

B. Inverse Displacement Analysis

The inverse displacement analysis is to determine the six active-joint displacements when the pose of the end-effector frame E with respect to the base frame B , i.e., $T_{B,E}$, is given. According to (1), when $T_{B,E}$ is given, the position vector of point P_i with respect to the base frame B , i.e., p_i^B ($i = 1, 2, 3$), can be directly computed. Hence, each of the three legs can be treated as an independent RPR type serial chain. Solving (4) with the known p_i^B , all joint displacements q_{bi} , q_{ci} and q_{di} can be determined. By projecting leg i onto a vertical plane, the prismatic joint displacement q_{ci} can be uniquely derived from the z coordinate of p_i^B . Then, projecting the leg onto the base plane, a planar RR type mechanism is obtained (Fig. 8). Using the given x and y coordinates of p_i^B , the two revolute-joint displacements q_{bi} and q_{di} can be determined, which have at most two sets of solutions. Therefore, at most eight sets of inverse displacement solutions exist for a complete 3RPRS PKM with DMA. Note that the inverse solution exist if and only if the inverse solutions exist for every individual leg.

IV. SINGULARITY ANALYSIS

A classical approach for singularity analysis is to formulate the instantaneous velocity relationship between the mobile platform and the active joints. Through rank analysis of the Jacobian matrices, three types of singularities, i.e., the inverse, forward, and combined singularities, can be identified [10], [30], which usually is not an easy task for a 6-DOF PKM. However, according to the specific design of the 3RPRS

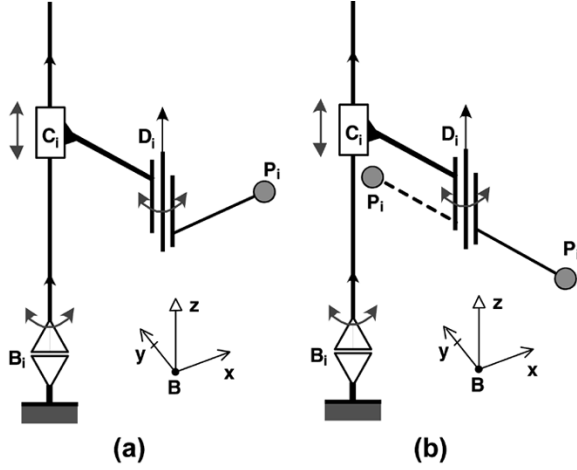


Fig. 9. Inverse singularity configurations.

PKM with DMA, a simple geometric approach can be employed to identify the singularity conditions. Formulation of instantaneous kinematics and Jacobian matrices can be avoided. Note that the mechanical interferences are ignored in the singularity analysis.

A. Inverse Singularity

The inverse singularity configuration refers to a specific robot configuration in which the moving platform loses one or more degrees of freedom, instantaneously. In other words, the active joint velocities cannot be determined from the given moving platform velocity at an inverse singularity configuration. For this specific PKM with DMA, each leg is connected to the moving platform through a spherical joint so that the point velocity of P_i can be uniquely determined with the given moving platform velocity. Hence, the inverse singularity analysis can be decoupled for each of the legs. Since each leg is a SCARA-like 3-DOF RPR type positional arm with an operational point P_i , $i = 1, 2, 3$, at the leg-end [Fig. 9(a)], its inverse singularity conditions are identical with those of a SCARA-type robot, i.e., the inverse singularity occurs if and only if the two links (lines) $C_i D_i$ and $D_i P_i$ in leg i are co-planar [Fig. 9(b)]. Geometrically, this singularity condition corresponds to the fully stretched or folded configurations. The complete 3RPRS PKM with DMA will suffer from inverse singularity if any leg is in an inverse singularity configuration [31].

B. Forward Singularity

The forward singularity configuration refers to a specific robot configuration in which the moving platform gains one or more degrees of freedom, instantaneously. Kinematically, it also means that there exist some nonzero vectors of the moving platform velocity when all of the active-joint velocities are equal to zero. In other words, if all of the active joints are completely locked, the moving platform will still possess infinitesimal motion in certain directions [30]. With such an insightful geometrical description, we can now investigate the forward singularity of the 3RPRS PKM with DMA in a simple and geometrical way. When the six active joints of the are rigidly locked, the 3RPRS PKM with DMA will be reduced to a 3RS closed-loop mechanism, as shown in Fig. 10. In a general case, it has no mobility and is a rigid structure. In some special configurations, however, the moving platform becomes unstable and instantaneous motions in certain directions may occur, which are exactly the cases of forward singularities. Based on the passive-joint-velocity method proposed in [17], whenever

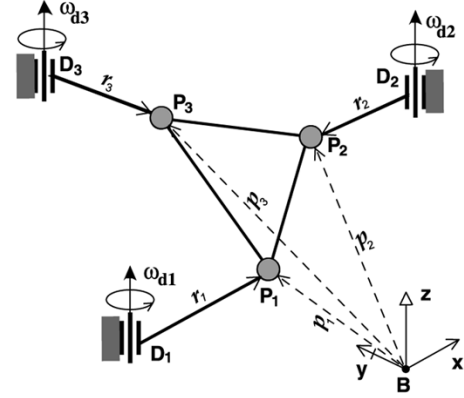


Fig. 10. Reduced 3RS parallel mechanism.

the moving platform has instantaneous motions, the following equation will always hold:

$$J_p \dot{q}_p = 0 \quad (8)$$

where $\dot{q}_p = (\dot{q}_{d1}, \dot{q}_{d2}, \dot{q}_{d3})$ represents the velocity of the three passive revolute joints, and matrix J_p is given by

$$J_p = \begin{bmatrix} p_{12}^T(r_1 \times w_{d1}) & -p_{12}^T(r_2 \times w_{d2}) & 0 \\ 0 & p_{23}^T(r_2 \times w_{d2}) & -p_{23}^T(r_3 \times w_{d3}) \\ -p_{31}^T(r_1 \times w_{d1}) & 0 & p_{31}^T(r_3 \times w_{d3}) \end{bmatrix}$$

with $p_{ij} = p_j - p_i$, $r_i = \overrightarrow{D_i P_i}$ and $w_{di} = (0, 0, 1)$ representing the unit directional vector of passive revolute joint D_i , ($i, j = 1, 2, 3, i \neq j$).

It can be depicted from (8) that the forward singularity will occur if and only if matrix J_p is singular so that $\dot{q}_p$ has nontrivial solutions. Hence, the complete and sufficient condition for the forward singularity of the 3RPRS PKM can be written as

$$\begin{aligned} \Delta_{J_p} &= p_{12}^T(r_1 \times w_{d1}) p_{23}^T(r_2 \times w_{d2}) p_{31}^T(r_3 \times w_{d3}) \\ &\quad - p_{12}^T(r_2 \times w_{d2}) p_{23}^T(r_3 \times w_{d3}) p_{31}^T(r_1 \times w_{d1}) \\ &= 0. \end{aligned} \quad (9)$$

Although (9) gives the mathematical expression of the forward singularity condition, it is still not geometrically clear. For this specific PKM with DMA, we can assume $p_i = (p_{ix}, p_{iy}, p_{iz})$ and $r_i = (r_{ix}, r_{iy}, 0)$. Through symbolic computation, we realized that Δ_{J_p} can also be written as

$$\Delta_{J_p} = \Delta_V \Delta_H \quad (10)$$

where $\Delta_V = (p_{12} \times p_{23}) \cdot w_{di}$ and Δ_H represents the determinant of matrix H , which is given by

$$H = \begin{bmatrix} r_{1y} & -r_{1x} & r_{1x}p_{1y} - r_{1y}p_{1x} \\ r_{2y} & -r_{2x} & r_{2x}p_{2y} - r_{2y}p_{2x} \\ r_{3y} & -r_{3x} & r_{3x}p_{3y} - r_{3y}p_{3x} \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (11)$$

It can also be symbolically verified that Δ_{J_p} is identical to the determinant of the 6×6 forward kinematics Jacobian matrix of the 3RPRS PKM with DMA. With (10), we can now geometrically conduct the forward singularity analysis. Since Δ_V represents the projection of the normal of the triangular plate $P_1 P_2 P_3$ onto the vertical direction w_{di} , Δ_V will take a value of zero if the triangular plate is perpendicular to the base plane, and in turn $\Delta_{J_p} = 0$. Therefore, the PKM with DMA will be always a forward singularity configuration when the tilting angle of the moving platform reaches 90° .

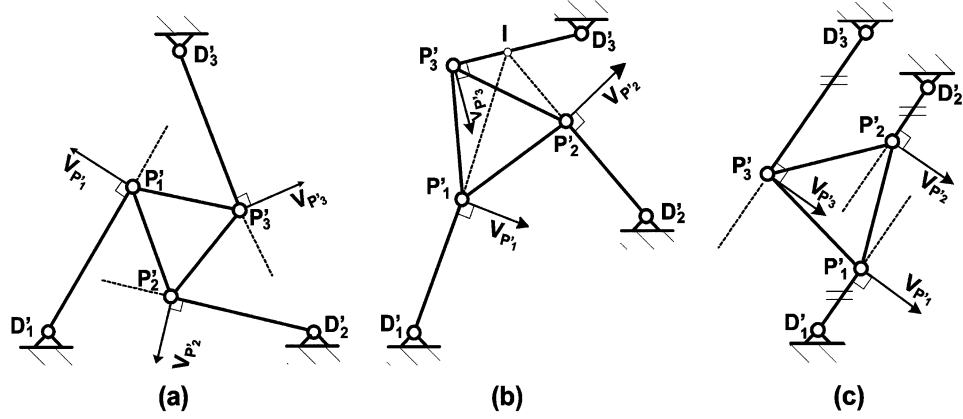


Fig. 11. Forward singularity configurations.

On the other hand, if the the triangular plate is not perpendicular to the base plane such that $\Delta_V \neq 0$, the forward singularity condition will only be dependent on the value of Δ_H , i.e., $\Delta_{JP} = 0$ if and only if $\Delta_H = 0$. It can be further verified that Δ_H is identical to the determinant of the forward kinematics Jacobian of the 3RRR planar PKM [31] obtained by projecting the 3RPRS PKM with DMA onto a horizontal plane. In other words, if the triangular plate is not perpendicular to the base plane, the forward singularity conditions can be derived from the projected 3RRR planar PKM, which has been well studied in [31] based on the concept of instantaneous centers. Here, we will only summarize the conclusions.

By projecting the 3RS closed-loop mechanism (Fig. 10) onto a horizontal plane, e.g., the base plane, a planar 3RR closed-loop mechanism (without mobility in a general configuration) can be obtained, as indicated in Fig. 11(a). Based on the method proposed in [31], it can be concluded that the forward singularity will occur if and only if the three projected links D_1P_1 , D_2P_2 , and D_3P_3 either intersect at a point [Fig. 11(b)] or are parallel to each other [Fig. 11(c)]. If the three links intersect at a point I as shown in Fig. 11(b), the projected moving platform can produce an infinitesimal rotation about the instantaneous center I . If the three links are parallel to each other as shown in Fig. 11(c), the projected moving platform can produce an infinitesimal translation relative to the base such that the instantaneous center lies at infinity. Otherwise, the moving platform and the base have an infinite number of instantaneous centers of zero velocity so that the moving platform is stationary with respect to the base.

Now, let us revert to the original 3RPRS PKM (Fig. 7). The revolute joint axis $\hat{s}_{di}$ and the center of spherical joint P_i of leg i can define a vertical plane. Hence, three such vertical planes can be formed. The forward singularity occurs if and only if the three vertical planes intersect at the same vertical line or they are parallel to each other. Moreover, if the moving platform is perpendicular to the base plane, it is always a forward singularity configuration.

C. Combined Singularity

The combined singularity configuration refers to a specific robot configuration in which the moving platform simultaneously gains and loses one or more degrees of freedom. Hence, the combined singularity occurs if and only if both forward and inverse singularities simultaneously occur. In other words, for a specific robot configuration, if the geometrical conditions for both forward and inverse singularities are simultaneously satisfied, it is a combined singularity configuration (Fig. 12). It is noted that the combined singularity is subject to strict conditions and can be avoided by proper dimension design of the mechanisms [10], [31]. For the specific PKM being developed, there are no combined singularities in the workspace.

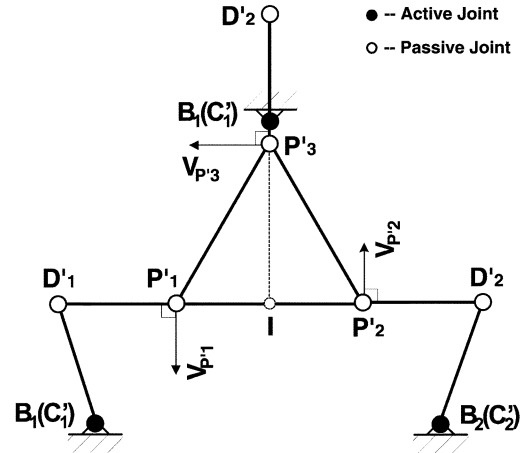


Fig. 12. Combined singularity configuration.

V. WORKSPACE ANALYSIS

As the complete 6-D workspace of the 3RPRS PKM with DMA is difficult to present, the major subject of this section is to determine the subsets of the workspace. For ease of implementation, a geometrical method is proposed to determine the constant-orientation workspace, i.e., the reachable workspace of the PKM with a constant moving-platform orientation. This method is based on the “intersection” technique proposed by Gosselin [18], in which the constant-orientation workspace is given by the intersection of individual ones generated by each of the three legs. Note that the mechanical interferences are neglected in workspace analysis.

In (1), the pose of the mobile platform frame $T_{B,E} \in SE(3)$ has the form of

$$T_{B,E} = \begin{bmatrix} R_{B,E} & p_{B,E} \\ 0 & 1 \end{bmatrix} \quad (12)$$

where $R_{B,E} \in SO(3)$ and $p_{B,E} \in \mathbb{R}^{3 \times 1}$ are the orientation and position of the moving platform frame relative to the base frame, respectively. Equation (1) thus can be rewritten as

$$p_{B,E} = p_i^B - R_{B,E} p_i^E \\ = p_i^B + p_{iR} \quad (13)$$

where $p_{iR} = -R_{B,E} p_i^E = (x_{iR}, y_{iR}, z_{iR}) \in \mathbb{R}^{3 \times 1}$.

Equation (13) indicates that, for a given moving platform orientation, its position $p_{B,E}$ can be determined by translating leg-end point P_i 's position through a constant vector p_{iR} . In other words, the constant-orientation workspace generated by leg i is a translation of point

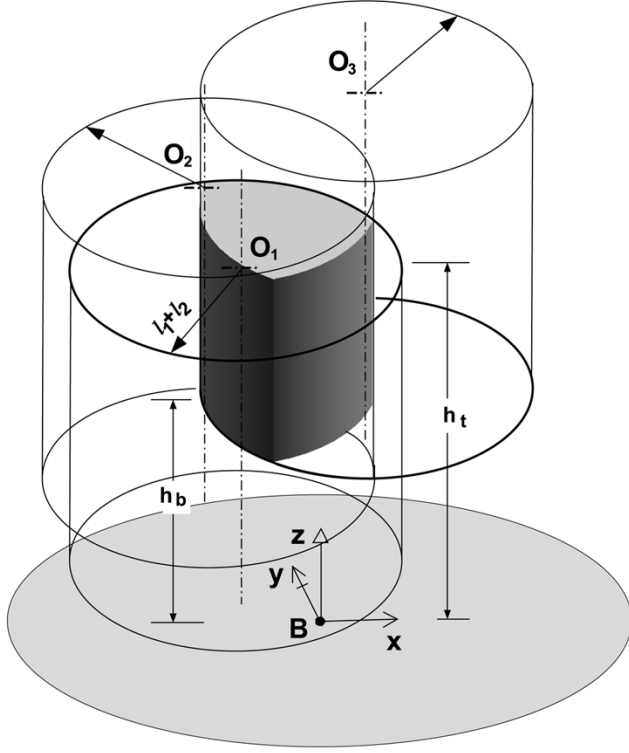


Fig. 13. Workspace of the 3RPRS PKM with DMA.

P_i 's reachable workspace. However, the motion of the moving platform is constrained by the three legs. The constant-orientation workspace of the 3RPRS PKM with DMA, therefore, is the intersection of the three individual reachable workspaces respectively generated by the three legs.

According to (4) and the specific RPR type leg design, the reachable workspace of the leg-end point P_i is a solid cylinder of radius $l_1 + l_2$, if there is no joint limitations for the two revolute joints ($\hat{s}_{bi}$ and $\hat{s}_{di}$) and $l_1 \leq l_2$, where l_1 and l_2 represent the lengths of links $C_i D_i$ and $D_i P_i$, respectively (Fig. 7). Furthermore, its center line is perpendicular to the base plane and passing through B_i . Hence, the reachable workspace of the leg-end point P_i can be mathematically expressed as

$$(x - x_{bi})^2 + (y - y_{bi})^2 \leq (l_1 + l_2)^2$$

$$h_{i \min} \leq z \leq h_{i \max} \quad (14)$$

where $h_{\min}$ and $h_{\max}$ represent the minimum and maximum z coordinates of point P_i , respectively. They can be readily computed from (4) with minimum and maximum prismatic joint displacements. Since the translation vector $p_{iR} = (x_{iR}, y_{iR}, z_{iR})$ can be computed from the given orientation of the moving platform, the constant-orientation workspace generated by leg i , i.e., the shifted reachable workspace, will be written as

$$(x - x_{bi} - x_{iR})^2 + (y - y_{bi} - y_{iR})^2 \leq (l_1 + l_2)^2$$

$$h_{i \min} + z_{iR} \leq z \leq h_{i \max} + z_{iR} \quad (15)$$

Note that the center line of the shifted solid cylinder, O_i ($i = 1, 2, 3$), is always perpendicular to the base plane, as shown in Fig. 13. Using the boundary conditions given in (15), the constant-orientation workspace can be determined, which is given by the intersection portion of the three shifted solid cylinders (Fig. 13). The height of the workspace can be computed from the z coordinates of the top and bottom surfaces of the three cylinders, while the cross-section area of the workspace can be

determined through projecting the three cylinders onto the base plane. It is apparent that the closer the three center lines and the larger the workspace height, the larger the constant-orientation workspace. The workspace analysis algorithm can be implemented to the workspace design and optimization of the 3RPRS PKM with DMA.

VI. SUMMARY

A new 6-DOF 3RPRS PKM with DMA has been designed and rapidly constructed with the standard joint modules and the customized links and moving plates. Such a PKM has the advantages of decoupled-motion characteristics, simple kinematics, large cylindrical reachable workspace, and high stiffness in the vertical direction. Hence, it is suitable for a variety of heavy parts assembly and light machining tasks. Based on the DMA, an effective projection approach has been proposed for the displacement, singularity, and workspace analysis. The displacement analysis shows that at most 12 sets of forward kinematic solutions and eight sets of inverse kinematic solutions exist for the 6-DOF 3RPRS PKM with DMA. Necessary and sufficient conditions for the inverse, forward, and combined singularity configurations are geometrically identified without formulating the Jacobian matrices. A computationally efficient geometrical algorithm has been formulated for the constant-orientation workspace analysis. In brief, the major contributions of this paper are twofold: the novel design of a 6-DOF PKM with DMA and the formulation of effective kinematics analysis algorithms for the PKM. Future work will be focused on dimension optimization of this 6-DOF 3RPRS PKM with DMA.

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Smooth Motion Generation for Unicycle Mobile Robots Via Dynamic Path Inversion

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Abstract—A new motion-generation approach is proposed for wheeled mobile robots described by the unicycle kinematic model. This approach permits the generation of smooth continuous-acceleration controls using a dynamic path-inversion procedure that exploits the concept of G^3 -paths, i.e., Cartesian paths with third-order geometric continuity (both the curvature function and its derivative, with respect to the arc length, are continuous). The exposed steering method is well suited to be adopted for the robot's iterative steering within a supervisory control architecture for sensor-based autonomous navigation. A worked example illustrates the approach.

Index Terms—Dynamic path inversion, geometric continuity, smooth motion generation, unicycle mobile robots.

I. INTRODUCTION

The work of Dubins [1] and Reeds and Shepp [2] devoted to the planning with minimal length paths are well-known fundamental results in the field of nonholonomic motion generation of wheeled mobile robots and car-like vehicles (see the survey [3] by Laumond *et al.*). A known limit of the Dubins–Reeds–Shepp approach is that the resulting paths are composed by line segments and circular arcs and do not have overall continuous curvature. Therefore, swift high-performance maneuvers cannot be achieved. At the end of the 1980s, many authors [4]–[6] proposed several curve primitives to overtake this limit: in all of the cases, continuous-curvature paths, usually denoted as G^2 -paths, were generated. Subsequently, many authors have worked on planning schemes ensuring continuous-curvature paths [7], [8]. A recent approach to continuous-curvature motion generation was proposed in [9] where continuous-curvature paths were obtained by using a new primitive, the η -spline or quintic G^2 -spline. This primitive was adopted to achieve a straightforward approach for the iterative steering of vision-based autonomous vehicles.

In this paper, the discussion is focused on the generation of a smooth motion control for unicycle mobile robots (UMRs), i.e., wheeled mobile robots whose kinematics is described by the unicycle model. In more detail, it will be demonstrated that, if such robots are driven with smooth linear and angular velocity command signals, i.e., signals which are continuous with their derivatives, they generate G^3 -paths, i.e., paths for which both the curvature function and its derivative with respect to the arc length are continuous. It will be further demonstrated that, conversely, the UMRs can be driven along any G^3 -path by means of appropriately devised C^1 command signals. A possible control strategy is then proposed. Its main features are the following.

- 1) A dynamic path inversion algorithm is designed to synthesize the UMR inputs. This algorithm requires planning with G^3 -paths [10] using interpolating conditions at the path end-points.

Manuscript received July 9, 2003; revised December 19, 2003. This paper was recommended for publication by Associate Editor K. Lynch and Editor H. Arai upon evaluation of the reviewers' comments. This work was supported in part by MIUR Scientific Research Funds under the framework of a COFIN 2002 project. This paper was presented in part at the 2002 IEEE Intelligent Vehicles Symposium, Versailles, France, June 18–20, 2002.

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Digital Object Identifier 10.1109/TRO.2004.832827